# 5.2 Congruent Polygons

Geometry 5.2 · Student sheet

Name _____ Period ____ Date _____

**I can…**

- identify corresponding parts of congruent figures.
- use corresponding parts to find unknown values.
- use the Third Angles Theorem.

**Vocabulary** · fill in the missing word

**Corresponding parts** — a pair of sides or angles that have the same relative position in two _____ figures

**Congruent figures** — figures that have the same size and _____

## Part 1 · Discovery: matching triangles

Make two congruent triangles. Label them ABC and DEF so A matches D, B matches E, and C matches F.

| Matching angles | Matching sides |
|---|---|
| ∠A and ____ | $\overline{AB}$ and ____ |
| ∠B and ____ | $\overline{BC}$ and ____ |
| ∠C and ____ | $\overline{AC}$ and ____ |

## Part 2 · Notes and examples (fill in each note, then work the example)

**Congruence statements:** in △ABC ≅ △DEF, the order of the letters tells which parts _____.

### Example 1

△ABC ≅ △DEF. List all pairs of congruent corresponding parts.

**Corresponding parts** of congruent figures are _____.

### Example 2

△ABC ≅ △DEF. AB = 2x + 3 and DE = 11. m∠C = (5y + 10)° and m∠F = 60°. Find x and y.

### Example 3

In △PQR and △STU: ∠P ≅ ∠T, ∠Q ≅ ∠U, ∠R ≅ ∠S, $\overline{PQ} \cong \overline{TU}$, $\overline{QR} \cong \overline{US}$, $\overline{RP} \cong \overline{ST}$. Write a congruence statement.

**Third Angles Theorem:** if two angles of one triangle are congruent to two angles of another triangle, then the third angles are

_____.

**Example 4**

In △ABC and △DEF, m∠A = m∠D = 45° and m∠B = m∠E = 70°. Find m∠C and m∠F. What can you conclude?

## Part 3 · Partner practice

**Partner 1**

△JKL ≅ △XYZ. Which side corresponds to $\overline{KL}$? Which angle corresponds to ∠L?

**Partner 2**

△MNP ≅ △RST. MN = 3x − 4 and RS = 17. Find x.